Analog Transmission

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Digital-to-Analog conversion

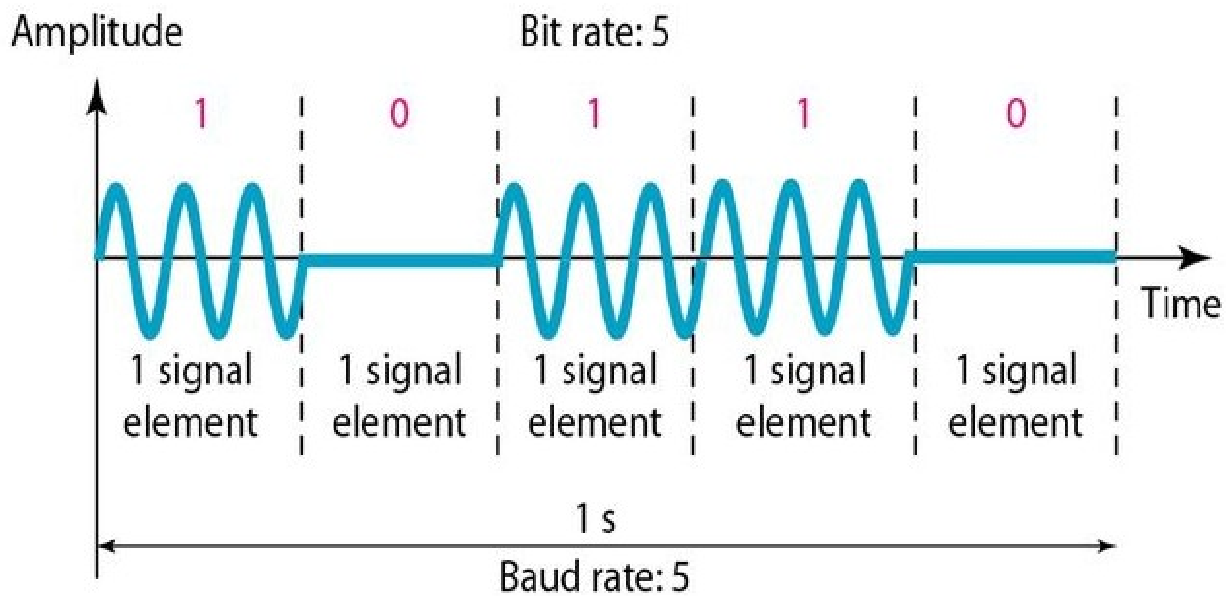
- It is the process of changing one of the characteristics of an analog signal based on the information in digital data
  - Amplitude shift keying (**ASK**)
  - Frequency shift keying (**FSK**)
  - Phase shift keying (**PSK**)

Bit rate vs. Baud rate

- In the analog transmission of digital data, the baud rate is less than or equal to the bit rate
  - $S = N \times 1/r$
- The value of r in analog transmission is $r = \log_2 L$, where L is the type of signal element

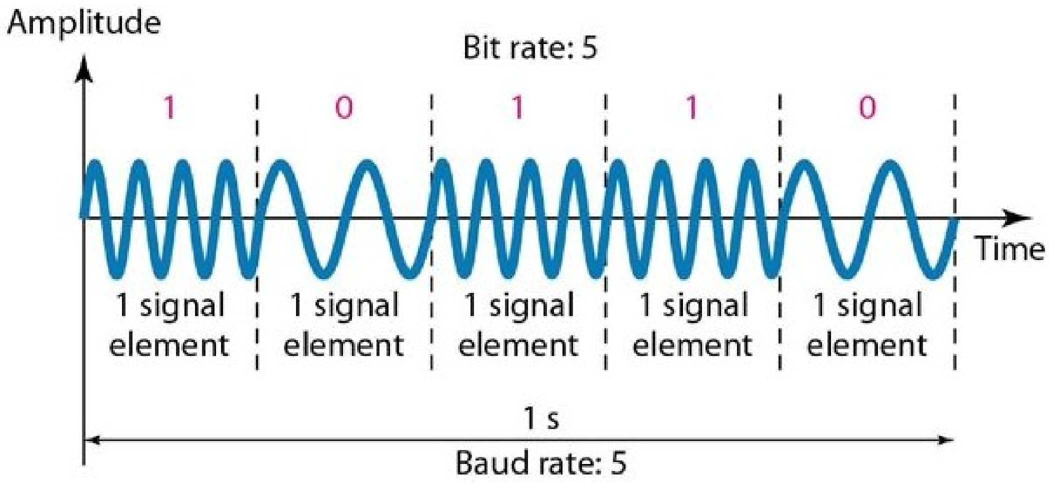
Amplitude Shift Keying (ASK)

- The amplitude of the carrier signal is varied to create signal elements
- Both frequency and phase remain constant
- ASK is normally implemented using only two levels called **Binary ASK (BASK)** or on-off keying
- The peak amplitude of one signal level is 0, the other is the same as the amplitude of the carrier frequency
- Bandwith is given by
- **B =(1 +d) x S**
- Where S is baud rate, and d is the factor for modulation and filtering process



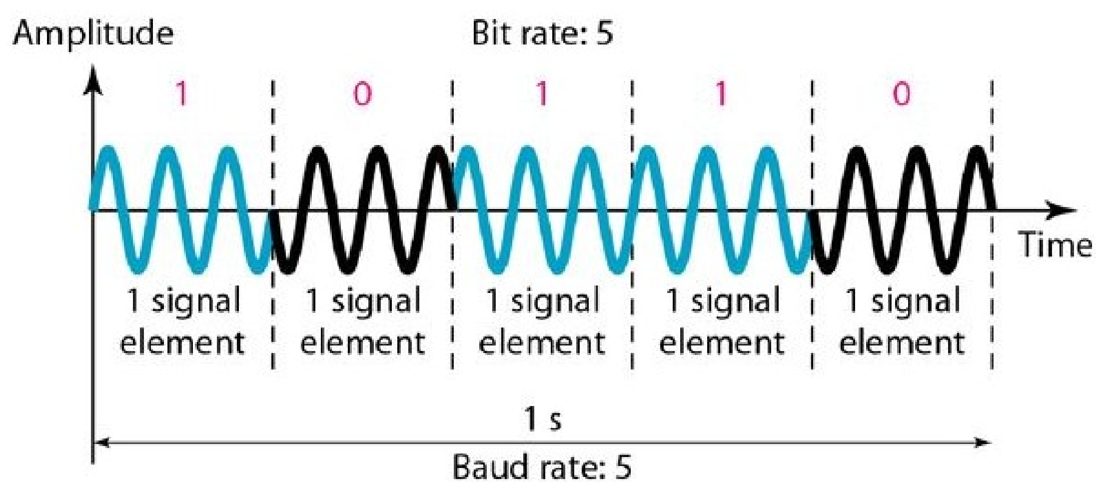
Frequency Shift Keying

- The frequency of the modulated signal is constant for the duration of one signal element, but changes for the next signal element if the data element changes
- **Binary FSK (BFSK)**
  - Consider two carrier frequencies
  - First carrier used if the data element is 0, otherwise second carrieer
  - If the difference between the two carrier frequencies is 2Δf, then the required bandwidth of BFSK is
    - **B=(1+d)xS+2Δf**
  - The minimum value of 2Δf should be at least S for the proper operation of modulation and demodulation



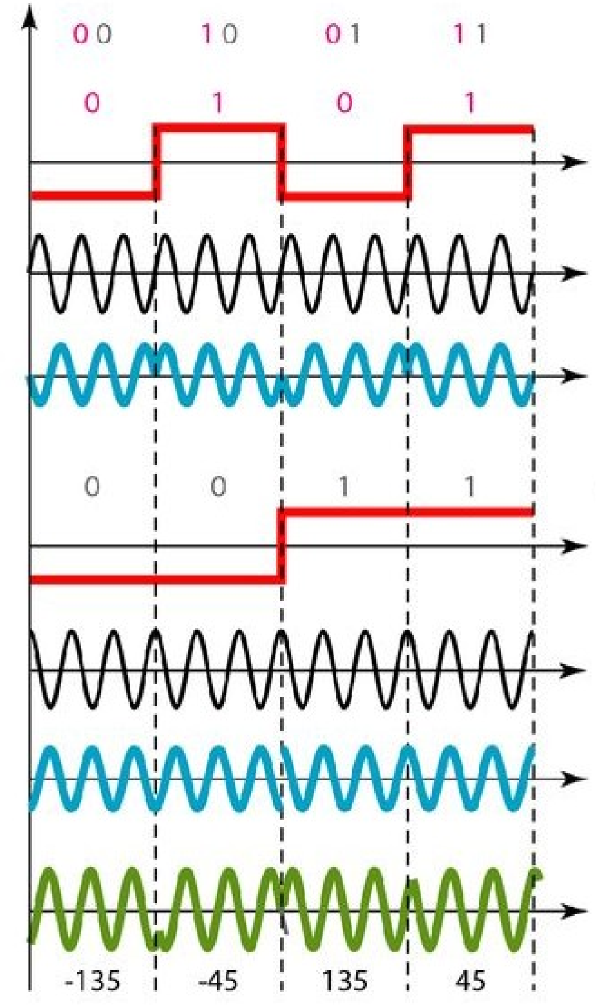
Phase Shift Keying

- The phase of the carrier is varied to represent two or more different signal elements
- **Binary PSK (BPSK)**
  - One signal element with a phase of 0°, and the other with a phase of 180°
  - Less susceptible to noise
  - The bandwidth is the same as that for BASK, but less than that for BFSK



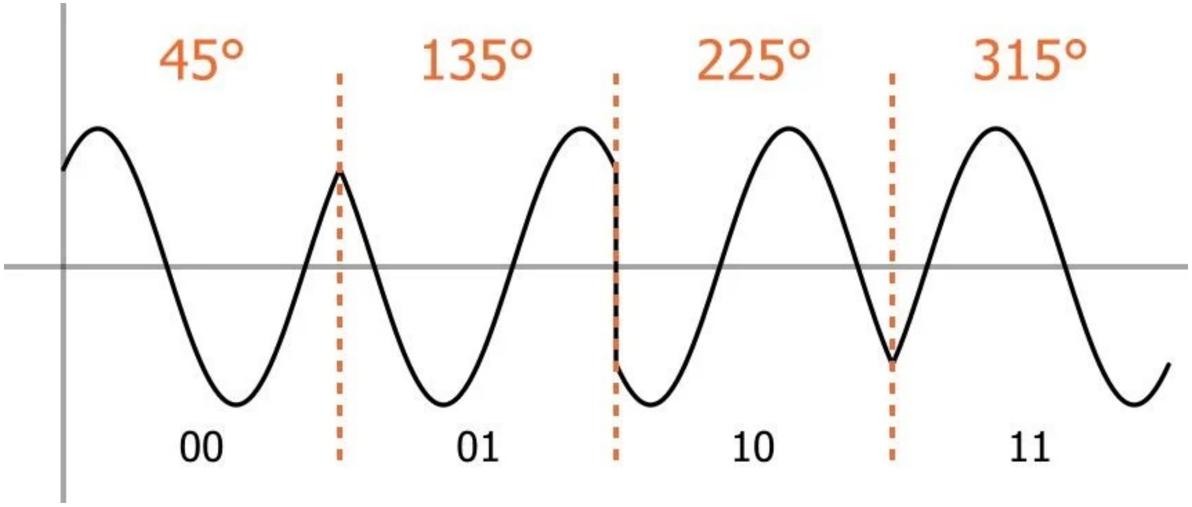
Quadrature PSK (QPSK)

- It uses two separate BPSK modulations; one is in-phase, the other quadrature (out-of-phase)
- If the duration of each bit in the incoming signal is T, the duration of each bit sent to the corresponding BPSK signal is 2T
- The two composite signals created by each multiplier are sine waves with the same frequency, but different phases



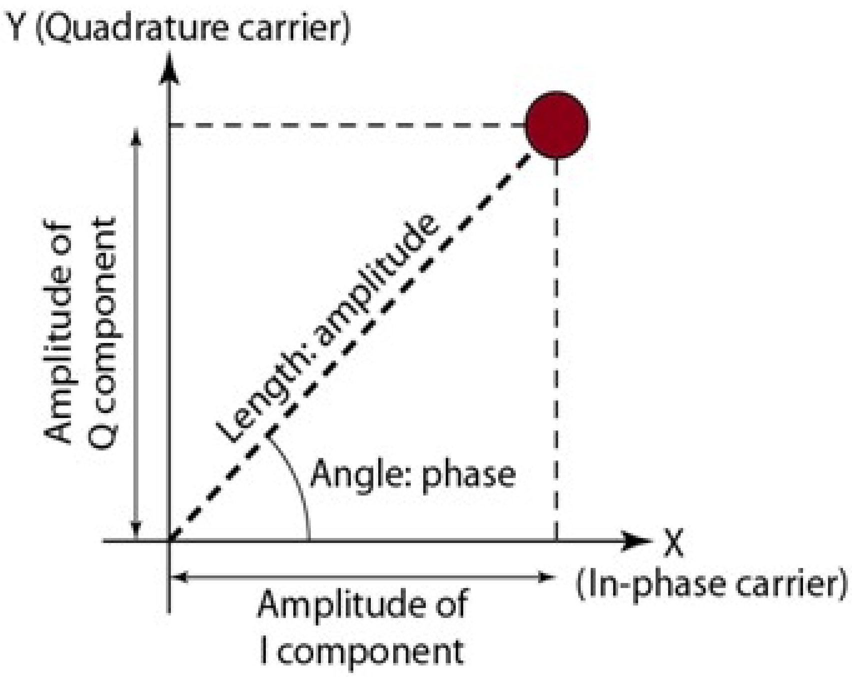
Quadrature PSK (QPSK)

- When they are added, the result is another sine wave, with one of four possible phases: 45°, -45° (315°), 135°, and -135°(225°)

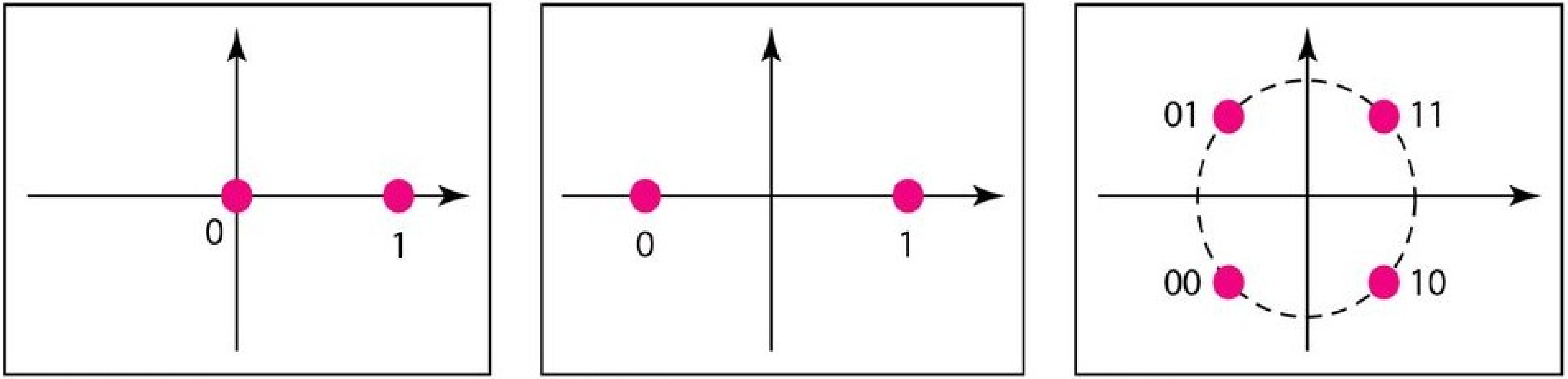


Constellatin Diagram

- A constellation diagram can help us define the amplitude and phase of a signal element, particularly when we are using two carriers (one in-phase and one quadrature)
- It is useful when we are dealing with multilevel ASK, PSK, or QAM



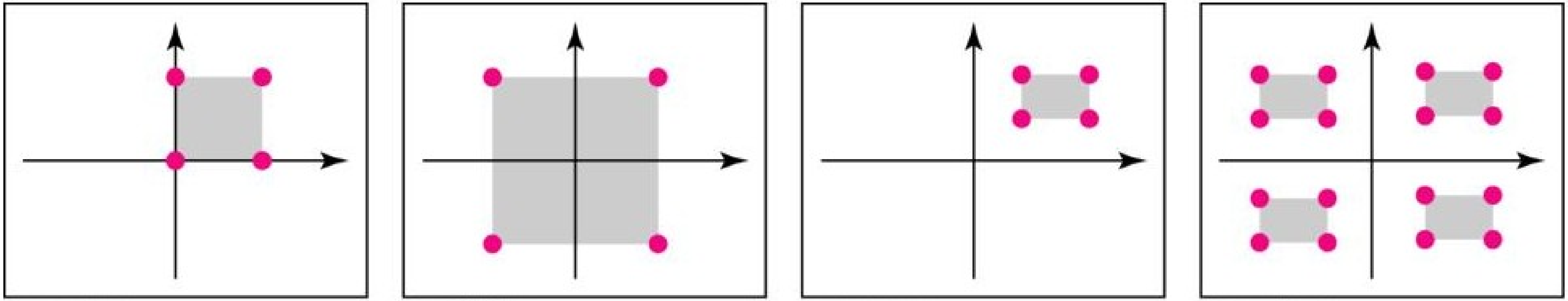
Constellation Diagram of ASK, PSK, and QPSK



a. ASK (OOK) b. BPSK c. QPSK

Quadrature Amplitude Modulation (QAM)

- The idea of using two carriers, one in-phase and the other quadrature, with different amplitude levels for each carrier .
- Quadrature amplitude modulation is a combination of ASK and PSK



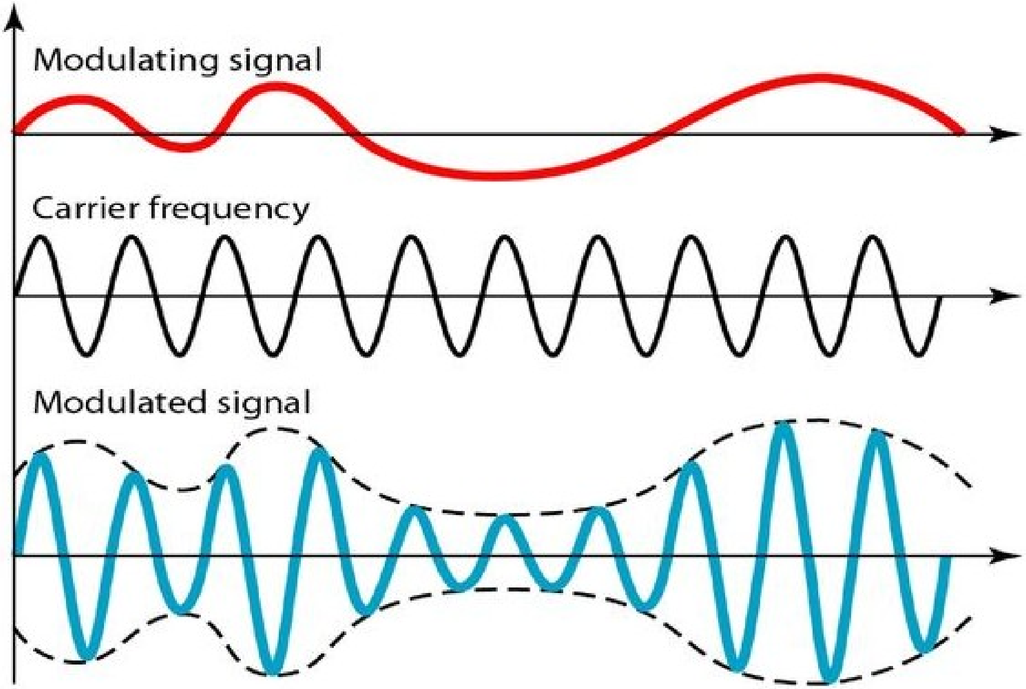
a. 4-QAM b. 4-QAM c. 4-QAM d. 16-QAM

Analog-to-Analog Conversion

- Modulation here is needed
  - If the medium is bandpass in nature or
  - If only a bandpass channel is available to us
  - Sometimes, low-pass signals need to be shifted, each to a different range
- 3 categories
  - Amplitude Modulation (AM)
  - Frequency Modulation (FM), and
  - Phase Modulation (PM)
- FM and PM are usually categorized together

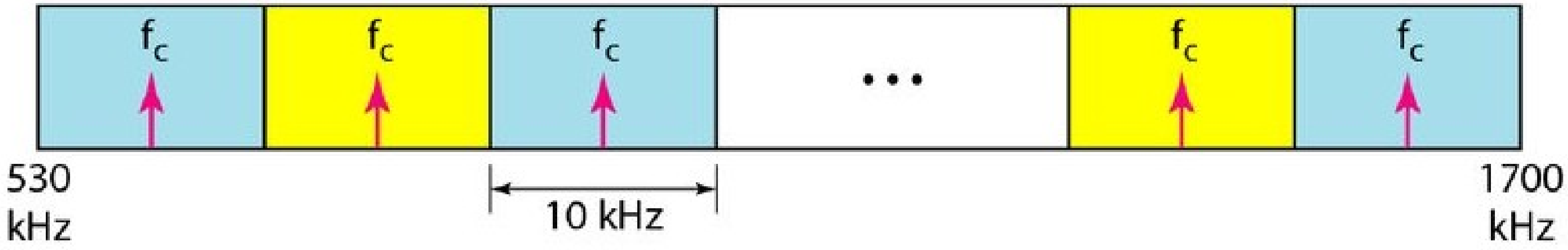
Amplitude Modulation

- The carrier signal is modulated so that its amplitude varies with the changing amplitudes of the modulating signal. The frequency and phase of the carrier remain the same



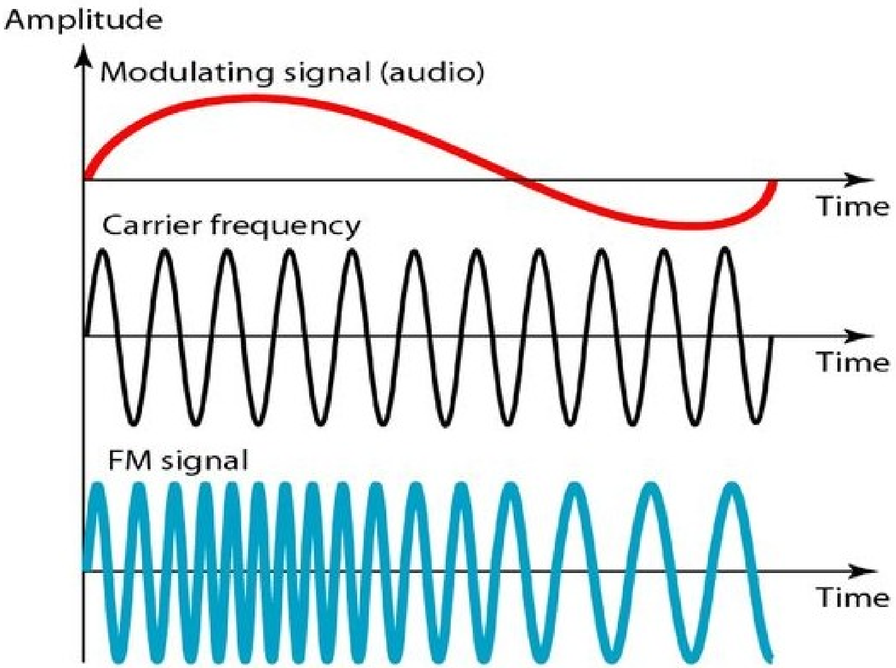
Bandwidth allocation of AM Radio

- The bandwidth of an audio signal (speech and music) is **usually 5 kHz**
- Therefore, an AM radio station needs a bandwidth of **10kHz**
- Federal Communications Commission (FCC) allows 10 kHz for each AM station.
- AM stations are allowed carrier frequencies anywhere between 530 and 1700 kHz (1.7 Mhz)
- Bandwidth separation by at least **10 kHz** to avoid interference



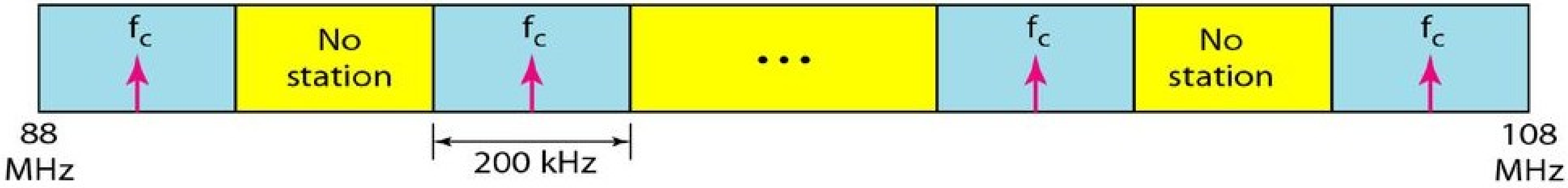
Frequency Modulation

- The frequency of the carrier signal is modulated to follow the changing voltage level (amplitude) of the modulating signal
- The peak amplitude and phase of the carrier signal remain constant
- As the amplitude of the information signal changes, the frequency of the carrier changes correspondingly
- The total bandwidth required for FM can be determined from the bandwidth of the audio signal by using $B_{FM} = 2(1 + \beta)B$, where β is the factor depends on modulation technique



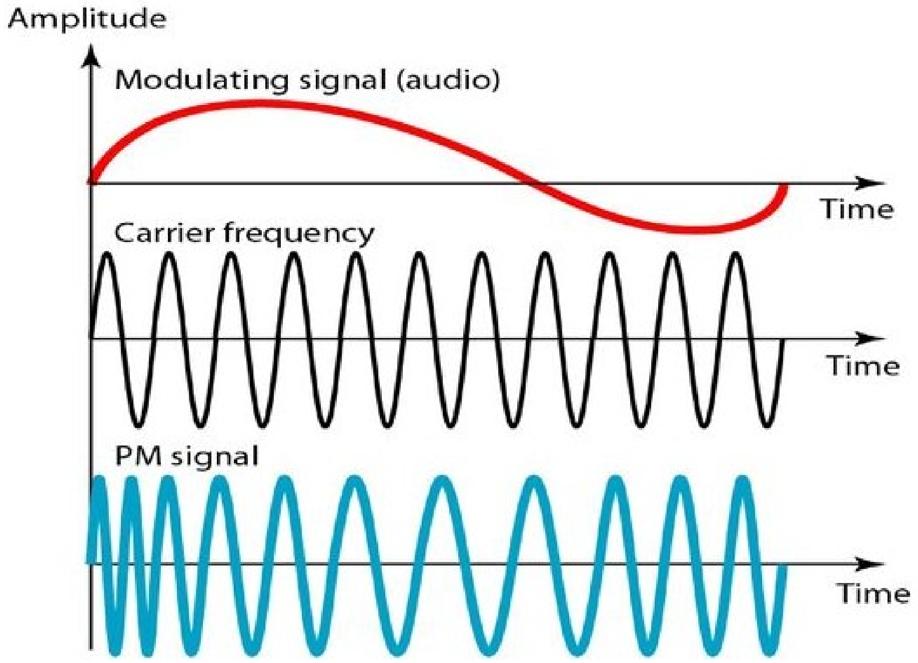
FM band allocation

- The bandwidth of an audio signal (speech and music) broadcast in stereo is almost **15 kHz**
- The FCC allows **200 kHz** (0.2 MHz) for each station (β = 4) with some extra guard band
- FM stations are allowed carrier frequencies anywhere between **88** and **108 MHz**
- Stations must be separated by **at least 200 kHz** to keep their bandwidths from overlapping
- To create even more privacy, the FCC requires that in a given area, only alternate bandwidth allocations may be used
- Given 88 to 108 MHz as a range, there are **100 potential PM bandwidths** in an area, of which **50 can operate at anyone time**



Phase Modulation

- The phase of the carrier signal is modulated to follow the changing voltage level (amplitude) of the modulating signal
- The peak amplitude and frequency of the carrier signal remain constant, but as the amplitude of the information signal changes, the phase of the carrier changes correspondingly



Next : Multiplexing